

03/09/26

Problem 1 :-

(a) Given:- a, b, c are 3 vectors $\in \mathbb{R}^3$

$$\therefore \text{let } \vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$

$$\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$$

$$\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$$

$$\text{Now, } b \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (b_2 c_3 - c_2 b_3) \hat{i} - (b_1 c_3 - b_3 c_1) \hat{j} + (b_1 c_2 - c_1 b_2) \hat{k}$$

$$a \cdot (b \times c) = a_1 (b_2 c_3 - c_2 b_3) + a_2 (b_3 c_1 - b_1 c_3) + a_3 (b_1 c_2 - c_1 b_2) \quad \text{--- (1)}$$

$$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2) \hat{i} - (a_1 b_3 - a_3 b_1) \hat{j} + (a_1 b_2 - b_1 a_2) \hat{k}$$

$$(a \times b) \cdot c = c_1 (a_2 b_3 - a_3 b_2) + c_2 (a_3 b_1 - a_1 b_3) + c_3 (a_1 b_2 - b_1 a_2)$$

$$= c_1 a_2 b_3 - c_1 a_3 b_2 + c_2 a_3 b_1 - c_2 a_1 b_3 + c_3 a_1 b_2 - c_3 b_1 a_2$$

Collecting like terms according to 'a' vector

$$= a_1 (b_2 c_3 - c_2 b_3) + a_2 (b_3 c_1 - c_3 b_1) + a_3 (b_1 c_2 - b_2 c_1) \quad \text{--- (i)}$$

which is the same expression as (i)

$$\therefore \boxed{a \cdot (b \times c) = (a \times b) \cdot c} \quad \text{is verified}$$

$$(b) \quad a \times (b \times c) = (a \cdot c) b - (a \cdot b) c \quad \text{As } a, b, c \in \mathbb{R}^3$$

$$\text{let } \vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$

$$\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$$

$$\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$$

$$\text{LHS} \Rightarrow b \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (b_2 c_3 - c_2 b_3) \hat{i} - (b_1 c_3 - c_1 b_3) \hat{j} + (b_1 c_2 - b_2 c_1) \hat{k}$$

$$a \times (b \times c) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_2 c_3 - c_2 b_3 & -b_1 c_3 + b_3 c_1 & b_1 c_2 - b_2 c_1 \end{vmatrix}$$

$$= (a_2 b_1 c_2 - a_2 b_2 c_1 - a_3 b_3 c_1 + a_3 b_1 c_3) \hat{i} - (a_1 b_1 c_2 - a_1 b_2 c_1 - a_3 b_2 c_3 + a_3 c_2 b_3) \hat{j} + (\cancel{a_1 b_3 c_1} - a_1 b_1 c_3 - a_2 b_2 c_3 + a_2 b_3 c_2) \hat{k}$$

- (1)

Now, $(a \cdot c) b$

$$= (a_1 c_1 + a_2 c_2 + a_3 c_3) b_1 \hat{i} + (a_1 c_1 + a_2 c_2 + a_3 c_3) b_2 \hat{j} + (a_1 c_1 + a_2 c_2 + a_3 c_3) b_3 \hat{k}$$

$$(a \cdot b) c = (a_1 b_1 + a_2 b_2 + a_3 b_3) c_1 \hat{i} + (a_1 b_1 + a_2 b_2 + a_3 b_3) c_2 \hat{j} + (a_1 b_1 + a_2 b_2 + a_3 b_3) c_3 \hat{k}$$

$$\begin{aligned}
 (a \cdot c) b - (a \cdot b) c &= \\
 (\cancel{a_1 b_1 c_1} + a_2 b_1 c_2 + a_3 b_1 c_3 - \cancel{a_1 b_1 c_1} - a_2 b_2 c_1 - a_3 b_3 c_1) \hat{i} \\
 + (a_1 b_2 c_1 + \cancel{a_2 b_2 c_2} + a_3 b_2 c_3 - a_1 b_1 c_2 - \cancel{a_2 b_2 c_2} - a_3 b_3 c_2) \hat{j} \\
 + (a_1 b_3 c_1 + a_2 b_3 c_2 + \cancel{a_3 b_3 c_3} - a_1 b_1 c_3 - a_2 b_2 c_3 - \cancel{a_3 b_3 c_3}) \hat{k} \\
 &= (a_2 b_1 c_2 - a_2 b_2 c_1 - a_3 b_3 c_1 + a_3 b_1 c_3) \hat{i} \\
 &\quad - (a_1 b_1 c_2 - a_1 b_2 c_1 + a_3 b_3 c_2 - a_3 b_2 c_3) \hat{j} \\
 &\quad + (a_1 b_3 c_1 - a_1 b_1 c_3 - a_2 b_2 c_3 + a_2 b_3 c_2) \hat{k}
 \end{aligned}$$

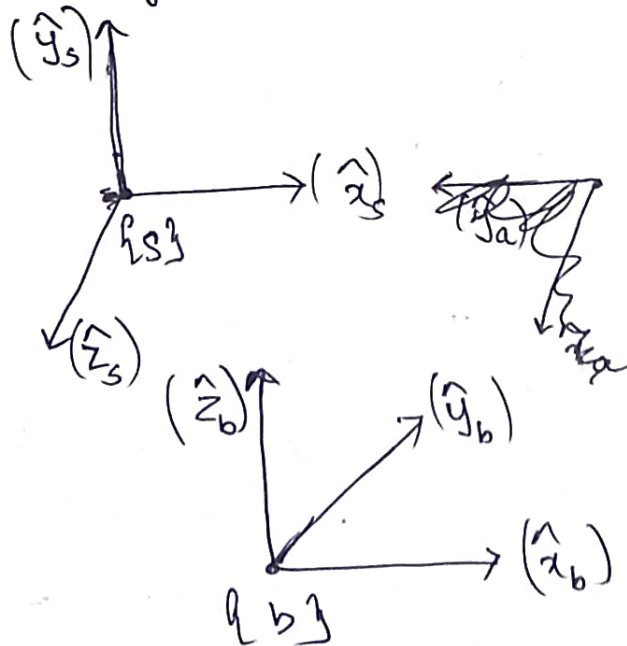
which is same as (1)

Hence,

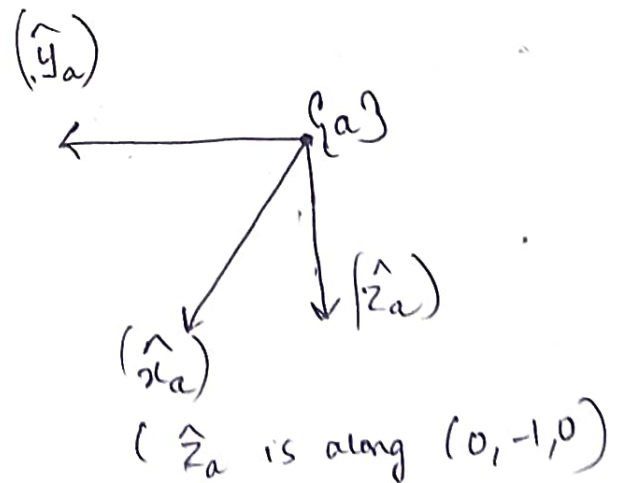
$$a \times (b \times c) = (a \cdot c) b - (a \cdot b) c \quad \text{is verified}$$

Problem 2

For visual clarity we will draw at 3 separate locations, even though it's actually at the same point, like origin is same for all 3. (the 3rd axis is drawn using right hand rule)



$(\hat{z}_b)$ is along $(0, 1, 0)$



$$R_{sa} = R_a^s$$

$$R_{sb} = R_b^s$$

$$\therefore R_{sa} = \begin{bmatrix} x_a^s & y_a^s & z_a^s \end{bmatrix} \quad R_{sb} = \begin{bmatrix} x_b^s & y_b^s & z_b^s \end{bmatrix}$$

$$= \begin{bmatrix} x_a \cdot x_s & y_a \cdot x_s & z_a \cdot x_s \\ x_a \cdot y_s & y_a \cdot y_s & z_a \cdot y_s \\ x_a \cdot z_s & y_a \cdot z_s & z_a \cdot z_s \end{bmatrix}$$

(similarly
\$R_{sb}\$

Using the coordinate frames we get:-

$$R_{sb} = \begin{bmatrix} x_b \cdot x_s & y_b \cdot x_s & z_b \cdot x_s \\ x_b \cdot y_s & y_b \cdot y_s & z_b \cdot y_s \\ x_b \cdot z_s & y_b \cdot z_s & z_b \cdot z_s \end{bmatrix}$$

$$R_{sa} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_{sb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

(c) We know from lecture notes (proved there)
if 0 & 1 are 2 frames

$$R_1^0 = (R_0^1)^T = (R_0^1)^{-1}$$

$$R_{sb}^{-1} = R_{bs} = (R_{sb})^T$$

$$R_{sb}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

(Taking the transpose)

which is \$R_{bs}\$

$$\therefore \text{each which is } R_s^b = \begin{bmatrix} x_s^b & y_s^b & z_s^b \end{bmatrix}$$

& from fig, this evaluates to -

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \text{ which is same as } R_{sb}^{-1} \text{ which proves our result.}$$

(d) ~~use~~ if $\bar{P}_s, \bar{P}_a, \bar{P}_b$ are the position vectors of the same point in the 3 frames, then,

we can say

$$\bar{P}_s = R_b^s \bar{P}_b$$

$$\bar{P}_a = R_b^a \bar{P}_b \quad \xrightarrow{\text{Substitute}}$$

$$\therefore \bar{P}_s = R_a^s R_b^a \bar{P}_b$$

comparing,

$$R_b^s = R_a^s R_b^a$$

$$(R_{ab} = R_b^a)$$

$\therefore$ ~~Take~~ Premultiply by $(R_a^s)^{-1}$
(since $R_a^s \in SO_3$ inverse exists)

$$\therefore R_b^a = (R_a^s)^{-1} R_b^s = I R_b^a$$

$$\therefore R_{ab} = (R_{sa})^{-1} R_{sb}$$

$$R_{ab} = R_{as} R_{sb}$$

Also

$$R_{ab} = (R_{sa})^T R_{sb}$$

$$\therefore R_{ab} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_{ab} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

From drawing,

$$R_{ab} = \begin{bmatrix} x_b^a & y_b^a & z_b^a \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

which is same as earlier hence verified

(e) $R_1 = R_{sa} R$

since $R = R_{sb}$ but its transformation operator
of rotation about $\hat{x}$ by -90°
for this elementary rotation about x $\theta = -90^\circ$

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\therefore R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

(which is
indeed
 R_{sb})

$$R_1 = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Rotation about $\hat{x}_S$ axis

corresponds $(-)$ rotation about $\hat{y}_a$ axis

(as they are collinear)
but dirn is opposite

$\therefore$ Consider rotation by -90° about $\hat{x}_S$ axis

$$\therefore R_3 = \begin{bmatrix} \cos 0 & 0 & \sin 0 \\ 0 & 1 & 0 \\ -\sin 0 & 0 & \cos 0 \end{bmatrix}$$

which $+90^\circ$ about $\hat{y}_a$ axis

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

which indeed matches an original earlier R_1

~~$\therefore R_1 = R_{sa} R$ indeed corresponds to rotation of R_{sa} by -90° about world fixed $\hat{x}_S$ axis.~~

if body fixed :-

$$R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 0 & -\sin 0 \\ 0 & \sin 0 & \cos 0 \end{bmatrix} \xrightarrow{-90^\circ} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

which does not match R_1

$$R_2 = R R_{sa}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{sa} \cdot R' = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is same as R_2

$$R_{sa} \cdot R'' = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

which is same as R_1

(We multiply by R_{sa} bcz R_1 & R_2 are final orientations but R' & R'' are rotation matrices $\rightarrow$ so multiply by R_{sa} so we can check final orientations)

$R'' \rightarrow$ body fixed

$R' \rightarrow$ world fixed

$\therefore$ From above, we can say,

R_1 corresponds to rotation of R_{sa} by -90° about the body fixed, x_a axis
 R_2 corresponds to rotation of R_{sa} by -90° about the world fixed x_s axis

$$(f) \quad \cancel{P_s} = \cancel{R_{sb}} \cancel{P_b}$$

$$P_s = R_{sb} P_b$$

$$\therefore P_s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$P_s = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

$$P_s = (1, 3, -2)$$

← s frame

$$(g) \quad P'' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad R_{sb}^T = R_{bs}$$

$$P'' = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

$$P'' = (1, -3, +2)$$

← P_b

This is multiplication with a rotation matrix where the physical location of Point P does not change but the coordinate frame changes, i.e., changing coordinates from {s} frame to {b} frame

$$P' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$P' = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

$$P' = (1, +3, -2)$$

~~The same thing holds for~~
~~The only diff~~

This is moving the location without changing the reference frame of the representation

(h) ω is still a vector so rotations apply for it as well

$$\therefore \omega_s = (3, 2, 1)$$

$$\omega_a = R_{as} \omega_s$$

$$\omega_a = (R_{sa})^T \omega_s$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}$$

$$\boxed{\omega_a = (1, -3, -2)}$$

Problem 3:-

Given $P = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{2}} \right) = \bar{P}_s$

let fixed frame be s

let $P \rightarrow P_1 \rightarrow P_2 \rightarrow P'$

$$P_1 = R_{x,\theta} P$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{6} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\theta = 30^\circ$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -1/2 \\ 0 & 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{6} \\ 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{3} \\ -\frac{1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{6}} + \frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{2} \\ 1/\sqrt{6} \end{bmatrix}$$

$$P_2 = R_{y, \phi} P_1$$

$$\phi = 135^\circ$$

$$= \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{2} \\ 1/\sqrt{6} \end{bmatrix}$$

$$= \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{2} \\ 1/\sqrt{6} \end{bmatrix}$$

$$= \begin{bmatrix} -1/\sqrt{6} + \frac{1}{\sqrt{12}} \\ -1/\sqrt{2} \\ -\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{12}} \end{bmatrix}$$

$$P' = R_{z, \psi} P_2$$

$$\psi = -120^\circ$$

$$= \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{6} + \frac{1}{\sqrt{12}} \\ -1/\sqrt{2} \\ -1/\sqrt{6} - \frac{1}{\sqrt{12}} \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{6} + \frac{1}{\sqrt{12}} \\ -1/\sqrt{2} \\ -1/\sqrt{6} - \frac{1}{\sqrt{12}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2\sqrt{6}} - \frac{1}{2\sqrt{12}} - \frac{\sqrt{3}}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{6}} - \frac{\sqrt{3}}{2\sqrt{12}} + \frac{1}{2\sqrt{2}} \\ -\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{12}} \end{bmatrix}$$

$$\frac{1}{2\sqrt{2}} - \frac{1}{4} + \frac{1}{2\sqrt{2}}$$

$$P' = \begin{bmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{12}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{4} \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{12}} \end{bmatrix}$$

(b) For $P' = R P$
 We saw that the consecutive rotations can be grouped into a single rotation by multiplying the consecutive rotation matrices (that too premultiply since these are rotations about 'fixed' frames).

proof:- $P' = R P$
 $P_1 = R_{x,\theta} P$
 $P_2 = R_{y,\phi} P_1$
 $P' = R_{z,\psi} P_2$

Substituting

$$P' = \underbrace{R_{z,\psi} R_{y,\phi} R_{x,\theta}}_R P$$

$$R = \begin{bmatrix} -\frac{1}{2} & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -1/2 \\ 0 & 1/2 & \sqrt{3}/2 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} +\frac{1}{2}\sqrt{2} & \frac{\sqrt{3}}{2} & -\frac{1}{2}\sqrt{2} \\ \frac{3}{2\sqrt{2}} & -\frac{1}{2} & -\frac{\sqrt{3}}{2}\sqrt{2} \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$R = \begin{bmatrix} \frac{1}{2}\sqrt{2} & \frac{3}{4} - \frac{1}{4\sqrt{2}} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4\sqrt{2}} \\ \frac{3}{2\sqrt{2}} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4\sqrt{2}} & \frac{1}{4} - \frac{3}{4\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{2\sqrt{2}} & -\frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix}$$

is the required rotation matrix

Problem 4:

To prove - $R_a^b R_b^c = R_a^c$

$$\{a\} = \{\hat{x}_a, \hat{y}_a, \hat{z}_a\} \quad \{b\} = \{\hat{x}_b, \hat{y}_b, \hat{z}_b\}$$

$$\{c\} = \{\hat{x}_c, \hat{y}_c, \hat{z}_c\}$$

$$\text{Also } \rightarrow \hat{x}_b = r_{11}\hat{x}_a + r_{21}\hat{y}_a + r_{31}\hat{z}_a$$

$$\text{for } \{b\} \quad \hat{y}_b = r_{12}\hat{x}_a + r_{22}\hat{y}_a + r_{32}\hat{z}_a$$

$$\text{and } \hat{z}_b = r_{13}\hat{x}_a + r_{23}\hat{y}_a + r_{33}\hat{z}_a$$

& which can be written as

$$\begin{bmatrix} \hat{x}_b \\ \hat{y}_b \\ \hat{z}_b \end{bmatrix} = \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix} \begin{bmatrix} \hat{x}_a \\ \hat{y}_a \\ \hat{z}_a \end{bmatrix}$$

Similarly for $\{c\}$

$$\begin{bmatrix} \hat{x}_c \\ \hat{y}_c \\ \hat{z}_c \end{bmatrix} = \begin{bmatrix} s_{11} & s_{21} & s_{31} \\ s_{12} & s_{22} & s_{32} \\ s_{13} & s_{23} & s_{33} \end{bmatrix} \begin{bmatrix} \hat{x}_b \\ \hat{y}_b \\ \hat{z}_b \end{bmatrix}$$

Now we know $R_a^b = \begin{bmatrix} x_a^b & y_a^b & z_a^b \end{bmatrix}$

$$= \begin{bmatrix} x_a \cdot x_b & y_a \cdot x_b & z_a \cdot x_b \\ x_a \cdot y_b & y_a \cdot y_b & z_a \cdot y_b \\ x_a \cdot z_b & y_a \cdot z_b & z_a \cdot z_b \end{bmatrix}$$

$$= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$\therefore$ From previous we can write,

$$\{b\} = R_a^b \{a\} \quad \text{--- (i)}$$

$\uparrow$ basis unit vectors in b frame $\uparrow$ basis unit vectors in a frame

Similarly.

$$R_b^c = \begin{bmatrix} x_b^c & y_b^c & z_b^c \end{bmatrix}$$

$$= \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix}$$

$$\{c\} = R_b^c \{b\}$$

Substitute (i) in here:

$$\{c\} = R_b^c R_a^b \{a\}$$

Also from analogy, we can also write

$$\{c\} = R_a^c \{a\}$$

comparing these 2 we get.

$$\boxed{R_a^c = R_b^c R_a^b} \quad \text{hence proved}$$

Or if we solve analytically,

Substituting in x_c ~~vector~~, y_c , z_c vector
(the b frame vectors)

we get.

$$\hat{x}_c = s_{11} r_{11} \hat{x}_a + s_{11} r_{21} \hat{y}_a + s_{11} r_{31} \hat{z}_a \\ + s_{21} r_{12} \hat{x}_a + s_{21} r_{22} \hat{y}_a + s_{21} r_{32} \hat{z}_a \\ + s_{31} r_{13} \hat{x}_a + s_{31} r_{23} \hat{y}_a + s_{31} r_{33} \hat{z}_a$$

$$\hat{y}_c = s_{12} r_{11} \hat{x}_a + s_{12} r_{21} \hat{y}_a + s_{12} r_{31} \hat{z}_a \\ + s_{22} r_{12} \hat{x}_a + s_{22} r_{22} \hat{y}_a + s_{22} r_{32} \hat{z}_a \\ + s_{32} r_{13} \hat{x}_a + s_{32} r_{23} \hat{y}_a + s_{32} r_{33} \hat{z}_a$$

$$\hat{z}_c = s_{13} r_{11} \hat{x}_a + s_{13} r_{21} \hat{y}_a + s_{13} r_{31} \hat{z}_a \\ + s_{23} r_{12} \hat{x}_a + s_{23} r_{22} \hat{y}_a + s_{23} r_{32} \hat{z}_a \\ + s_{33} r_{13} \hat{x}_a + s_{33} r_{23} \hat{y}_a + s_{33} r_{33} \hat{z}_a$$

$$\therefore \begin{bmatrix} \hat{x}_c \\ \hat{y}_c \\ \hat{z}_c \end{bmatrix} = \begin{bmatrix} s_{11} r_{11} + s_{21} r_{12} + s_{31} r_{13} \\ s_{12} r_{11} + s_{22} r_{12} + s_{32} r_{13} \\ s_{13} r_{11} + s_{23} r_{12} + s_{33} r_{13} \end{bmatrix}$$

$$\begin{bmatrix} \hat{x}_c \\ \hat{y}_c \\ \hat{z}_c \end{bmatrix} = \begin{bmatrix} s_{11} r_{11} + s_{21} r_{12} + s_{31} r_{13} & s_{11} r_{21} + s_{21} r_{22} + s_{31} r_{23} & s_{11} r_{31} + s_{21} r_{32} + s_{31} r_{33} \\ s_{12} r_{11} + s_{22} r_{12} + s_{32} r_{13} & s_{12} r_{21} + s_{22} r_{22} + s_{32} r_{23} & s_{12} r_{31} + s_{22} r_{32} + s_{32} r_{33} \\ s_{13} r_{11} + s_{23} r_{12} + s_{33} r_{13} & s_{13} r_{21} + s_{23} r_{22} + s_{33} r_{23} & s_{13} r_{31} + s_{23} r_{32} + s_{33} r_{33} \end{bmatrix} \begin{bmatrix} \hat{x}_a \\ \hat{y}_a \\ \hat{z}_a \end{bmatrix}$$

This is R_a^c
which transforms a frame to c frame

if

$$R_a^b R_b^c = R_a^c$$

then

$$R_a^c = \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix} \begin{bmatrix} s_{11} & s_{21} & s_{31} \\ s_{12} & s_{22} & s_{32} \\ s_{13} & s_{23} & s_{33} \end{bmatrix}$$

$$= \begin{bmatrix} r_{11}s_{11} + r_{21}s_{12} + r_{31}s_{13} & r_{11}s_{21} + r_{21}s_{22} + r_{31}s_{23} & r_{11}s_{31} + r_{21}s_{32} + r_{31}s_{33} \\ r_{12}s_{11} + r_{22}s_{12} + r_{32}s_{13} & r_{12}s_{21} + r_{22}s_{22} + r_{32}s_{23} & r_{12}s_{31} + r_{22}s_{32} + r_{32}s_{33} \\ r_{13}s_{11} + r_{23}s_{12} + r_{33}s_{13} & r_{13}s_{21} + r_{23}s_{22} + r_{33}s_{23} & r_{13}s_{31} + r_{23}s_{32} + r_{33}s_{33} \end{bmatrix}$$

which indeed matches the previous R_a^c

$\therefore$ It's proved,

$$\boxed{R_a^b R_b^c = R_a^c}$$

Problem 5

$$\text{let } R = \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix}$$

$$R \in SO_3$$

$$\bar{u}, \bar{v} \in \mathbb{R}^3$$

$$\text{let } \bar{v} = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$$

$$\bar{u} = u_1 \hat{i} + u_2 \hat{j} + u_3 \hat{k}$$

$\rightarrow$

$$V \times W = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

$$= (v_2 w_3 - w_2 v_3) \hat{i} - (v_1 w_3 - w_1 v_3) \hat{j} + (v_1 w_2 - v_2 w_1) \hat{k}$$

+ n

Consider the column vectors of R .

$$r_1 = \cancel{r_{12}} (r_{11}, r_{12}, r_{13})$$

$$r_2 = (r_{21}, r_{22}, r_{23})$$

$$r_3 = (r_{31}, r_{32}, r_{33})$$

$$\therefore R = [\bar{r}_1 \quad \bar{r}_2 \quad \bar{r}_3]$$

$$\therefore Rv = \begin{bmatrix} \bar{r}_1 \cdot v_1 \\ \vdots \end{bmatrix}$$

$$Rv = v_1 \bar{r}_1 + v_2 \bar{r}_2 + v_3 \bar{r}_3$$

Also

$$Rw = w_1 \bar{r}_1 + w_2 \bar{r}_2 + w_3 \bar{r}_3$$

$$\begin{aligned} (Rv) \times (Rw) &= v_1 w_1 (\bar{r}_1 \times \bar{r}_1) + v_1 w_2 (\bar{r}_1 \times \bar{r}_2) + \\ &\quad v_1 w_3 (\bar{r}_1 \times \bar{r}_3) + v_2 w_1 (\bar{r}_2 \times \bar{r}_1) + \\ &\quad v_2 w_2 (\bar{r}_2 \times \bar{r}_2) + v_2 w_3 (\bar{r}_2 \times \bar{r}_3) \\ &\quad + v_3 w_1 (\bar{r}_3 \times \bar{r}_1) + v_3 w_2 (\bar{r}_3 \times \bar{r}_2) + \\ &\quad v_3 w_3 (\bar{r}_3 \times \bar{r}_3) \end{aligned}$$

Now, we know,

Since $\bar{r}_1, \bar{r}_2, \bar{r}_3$ vectors are orthonormal

$$\therefore \bar{r}_1 \times \bar{r}_1 = \bar{r}_2 \times \bar{r}_2 = \bar{r}_3 \times \bar{r}_3 = 0 \quad (\text{self cross product is } 0)$$

$$\begin{aligned}\bar{r}_1 \times \bar{r}_2 &= \bar{r}_3 & \bar{r}_1 \times \bar{r}_3 &= -\bar{r}_2 \\ \bar{r}_2 \times \bar{r}_1 &= -\bar{r}_3 & \bar{r}_2 \times \bar{r}_3 &= \bar{r}_1 \\ \bar{r}_3 \times \bar{r}_1 &= \bar{r}_2 & \bar{r}_3 \times \bar{r}_2 &= -\bar{r}_1\end{aligned}$$

put all these

$$\begin{aligned}\therefore (\bar{r}_1 \times \bar{r}_2) \times (\bar{r}_1 \times \bar{r}_3) &= v_1 \omega_2 \bar{r}_3 - v_1 \omega_3 \bar{r}_2 - v_2 \omega_1 \bar{r}_3 \\ &\quad + v_2 \omega_3 \bar{r}_1 + v_3 \omega_1 \bar{r}_2 - v_3 \omega_2 \bar{r}_1 \\ &= (v_2 \omega_3 - v_3 \omega_2) \bar{r}_1 \\ &\quad + (v_3 \omega_1 - v_1 \omega_3) \bar{r}_2 \\ &\quad + (v_1 \omega_2 - v_2 \omega_1) \bar{r}_3\end{aligned}$$

$$= \begin{bmatrix} \bar{r}_1 & \bar{r}_2 & \bar{r}_3 \end{bmatrix} \begin{bmatrix} v_2 \omega_3 - v_3 \omega_2 \\ v_3 \omega_1 - v_1 \omega_3 \\ v_1 \omega_2 - v_2 \omega_1 \end{bmatrix}$$

Now, from earlier we saw,

$$\begin{aligned}\bar{v} \times \bar{\omega} &= (v_2 \omega_3 - v_3 \omega_2) \hat{i} \\ &\quad + (v_3 \omega_1 - v_1 \omega_3) \hat{j} \\ &\quad + (v_1 \omega_2 - v_2 \omega_1) \hat{k}\end{aligned}$$

$$\therefore \bar{v} \times \bar{\omega} = \begin{bmatrix} v_2 \omega_3 - v_3 \omega_2 \\ v_3 \omega_1 - v_1 \omega_3 \\ v_1 \omega_2 - v_2 \omega_1 \end{bmatrix}$$

$$\& \begin{bmatrix} \bar{r}_1 & \bar{r}_2 & \bar{r}_3 \end{bmatrix} = R$$

$\therefore$ its proved

$$\boxed{(\bar{r}_1 \times \bar{r}_2) \times (\bar{r}_1 \times \bar{r}_3) = R (\bar{v} \times \bar{\omega})}$$

(b) We know,

$$\cancel{\omega \times \bar{v}} \quad \bar{v} \times \bar{\omega} = \hat{v} \omega$$

Consider LHS first

$$R \hat{\omega} R^T$$

$$\text{Now } \hat{\omega}(v) = \omega \times v$$

if v is a vector.

but here R^T is a matrix

consider multiplication by a v vector to this

$$\therefore R \hat{\omega} R^T v = (R \hat{\omega} (R^T v))$$

$$\text{Now } R^T v \text{ is a vector}$$

$$\hat{\omega}(v) = \omega \times v$$

$$\therefore R \hat{\omega} R^T v = R (\bar{\omega} \times (R^T v))$$

Now from part a we know

$$\{R(\bar{v} \times \bar{\omega})\} = (R \bar{v}) \times (R \bar{\omega})$$

$$\therefore R \hat{\omega} R^T v = (R \bar{\omega}) \times \{R(R^T v)\}$$

matrix multiplication
is associative

$$\therefore R \hat{\omega} R^T v = (R \bar{\omega}) \times (R R^T) v$$

$$R R^T = R^T \Rightarrow R R^T = I \Rightarrow R R^T = I$$

$$\therefore R \hat{\omega} R^T v = (R \bar{\omega}) \times v \quad \leftarrow (R \bar{\omega}) \text{ is a } R^3 \text{ vector so compatible}$$

Now if $\bar{\omega} \times \bar{v}$ then it can be written as $\hat{\omega} \bar{v}$

$$R \hat{\omega} R^T \bar{v} = (\hat{R \bar{\omega}}) \bar{v}$$

Comparing we get

$$\boxed{R \hat{\omega} R^T = \hat{R \bar{\omega}}}$$

hence proved

(c) Now, $R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \end{bmatrix}$

Expand this

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\det R = r_{11} (r_{22} r_{33} - r_{23} r_{32}) - r_{12} (r_{21} r_{33} - r_{13} r_{32}) + r_{13} (r_{21} r_{32} - r_{12} r_{23})$$

Now $r_2 \times r_3$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{vmatrix}$$

$$= (r_{22} r_{33} - r_{23} r_{32}) \hat{i} - (r_{21} r_{33} - r_{31} r_{23}) \hat{j} + (r_{21} r_{32} - r_{31} r_{22}) \hat{k}$$

$$r_1^T (r_2 \times r_3) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \end{bmatrix} \begin{bmatrix} r_{22} r_{33} - r_{23} r_{32} \\ r_{31} r_{23} - r_{21} r_{33} \\ r_{21} r_{32} - r_{31} r_{22} \end{bmatrix}$$

$$= r_{11} (r_{22} r_{33} - r_{23} r_{32}) + r_{12} (r_{31} r_{23} - r_{21} r_{33}) + r_{13} (r_{21} r_{32} - r_{31} r_{22})$$

Rearrange and take r_{21} & r_{31} common

$$\begin{aligned} r_{11}^T (r_2 \times r_3) &= r_{11} (r_{22} r_{33} - r_{23} r_{32}) \\ &\quad - r_{21} (r_{12} r_{33} - r_{13} r_{32}) \\ &\quad + r_{31} (r_{12} r_{23} - r_{13} r_{22}) \\ &= \det R \end{aligned}$$

$$\boxed{\det R = r_{11}^T (r_2 \times r_3)}$$

(d) To prove if a subset 'S' is a vector space,

i) $0 \in S$

ii) If $A, B \in S$ then $A+B \in S$

iii) $A \in S$ then $cA \in S$

Given $SO(3)$ (the set of 3×3 skew symmetric matrices) — let the set be 'S'

To prove — it is a subspace)

i) $0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is indeed is a 3×3 skew symmetric matrix.

$\therefore \underline{\underline{0 \in S}}$

ii) Let $A = \begin{bmatrix} a & b & c \\ -b & d & e \\ -c & -e & f \end{bmatrix}$. $B = \begin{bmatrix} p & q & r \\ -q & s & t \\ -r & -t & u \end{bmatrix}$

$A, B \in S$ (skew symmetric 3×3)

$$A+B = \begin{bmatrix} a+p & b+q & c+r \\ -(b+q) & d+s & e+t \\ -(c+r) & -(e+t) & u+f \end{bmatrix}$$

which is a 3×3 skew symmetric matrix

$$\therefore A + B \in S$$

$$(iv) \quad \text{co}A \in S \quad \text{or } KA = \begin{bmatrix} ka & kb & kc \\ -kb & kd & ke \\ -kc & -ke & kf \end{bmatrix}$$

is a 3×3 skew symmetric matrix

$\therefore$ when $A \in S$ ~~so~~ $KA \in S$
(or KA)

All points are proved
so $so(3)$ is indeed a vector space.

[here ~~as~~ a, d, f must be 0]
for skew symmetric

(just written in general)

(Also P, S, u)

$\therefore$ there are only 3 unknowns, (b, c, e)
alt ~~the~~ the entire ~~sub~~ vector space
can be expressed ~~to~~ using these
3 unknowns, that's exactly what a
dimension is

$$\therefore \boxed{\dim(so(3)) = 3}$$

For basis, basis are the three ~~constants~~ components
of linear combination of which any $A \in S$
can be expressed like
for $\mathbb{R}^3$ if $\vec{a} = a\hat{i} + b\hat{j} + c\hat{k}$
($\hat{i}, \hat{j}, \hat{k}$ are the basis) (a, b, c) are unknowns

For any $A \in S$

$$A = a \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

basis

∴ Basis is

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

Problem 6: Given $SO(2)$ be the set of 2×2 orthogonal matrices with determinant equal to +1.

(a)

(a) To show,

$SO(2)$ can be identified with S^1 , the unit circle in $\mathbb{R}^2$. We know $S^1 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x^2 + y^2 = 1 \right\}$

Let $A \in SO(2)$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

→ since the columns are unit vectors and orthogonal to each other,

$$a^2 + c^2 = 1 \rightarrow a, c \text{ can be expressed in } S^1$$

$$b^2 + d^2 = 1 \rightarrow b, d \text{ can be expressed in } S^1$$

$$ab + cd = 0$$

$$\det A = ad - bc = 1$$

$$ab = -cd$$

$$a = -\frac{cd}{b}$$

$b \neq 0$

$$\therefore \underline{a = d}$$

$$-\frac{cd^2}{b} - bc = 1$$

$$-c(d^2 + b^2) = 1$$

$$b = -c$$

$$\therefore \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} -c \\ a \end{bmatrix}$$

$$A = \begin{bmatrix} a & -c \\ c & a \end{bmatrix}$$

$$a^2 + c^2 = 1$$

Also

$$(-c)^2 + (a)^2 =$$

$$a^2 + c^2 = 1$$

both columns can be expressed in S_1 using just 2 variables (a, c)

$\therefore SO(2)$ can be identified with S^1 , the unit circle in $\mathbb{R}^2$

Just choose any point on the circle its 2 coordinates can form an $SO(2)$ matrix, so identification possible

(b) $R \in SO(2)$ $\hat{\omega} \in so(2)$
To prove $R \hat{\omega} R^T = \hat{\omega}$

let $R = A = \begin{bmatrix} a & -c \\ c & a \end{bmatrix}$

Also $so(2)$ is a skew-symmetric matrix

So, it can be written as

$$\hat{\omega} = \begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix}$$

$$\begin{aligned} R \hat{\omega} &= \begin{bmatrix} a & -c \\ c & a \end{bmatrix} \begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix} \\ &= \begin{bmatrix} c\omega & a\omega \\ -a\omega & c\omega \end{bmatrix} \end{aligned}$$

$$(\hat{R}\hat{\omega})R^T = \begin{bmatrix} c\omega & a\omega \\ -a\omega & c\omega \end{bmatrix} \begin{bmatrix} a & c \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} \cancel{ac\omega} - \cancel{ac\omega} & c^2\omega + a^2\omega \\ -a^2\omega - c^2\omega & -\cancel{ac\omega} + \cancel{ac\omega} \end{bmatrix}$$

2

$$a^2 + c^2 = 1$$

$$\begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix}$$

which is
same as
 $\hat{\omega}$

∴ it's proved

$$\hat{R}\hat{\omega}R^T = \hat{\omega}$$